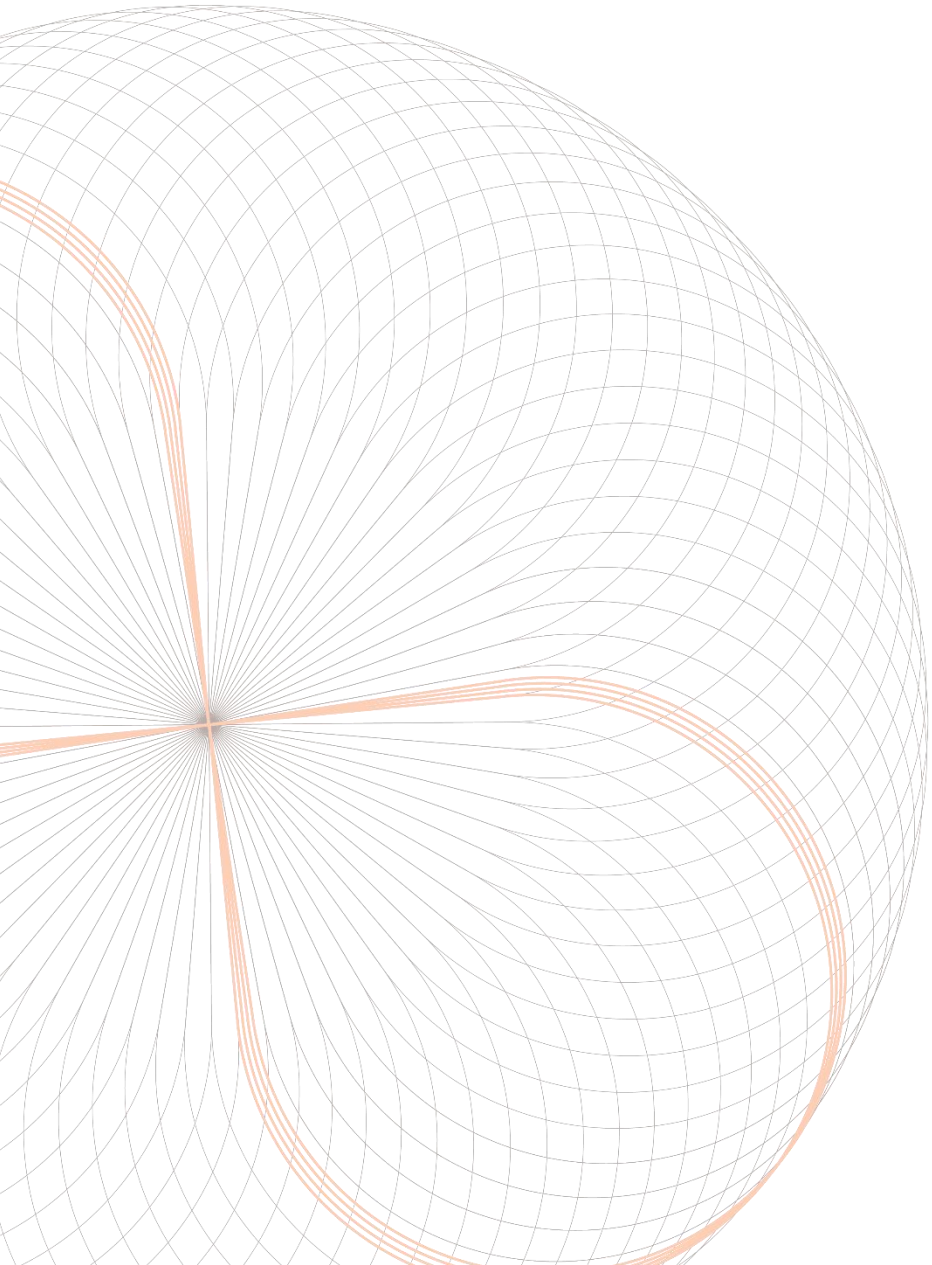
Margin and collateral
intelligence for derivatives.

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Cassini Technical Assessment

QA Automation Engineer

July 2025



Margin and collateral
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Assessment Details

The purpose of this assessment is to gain an understanding of your current knowledge base in test design, which is used daily as a QA Engineer at Cassini.

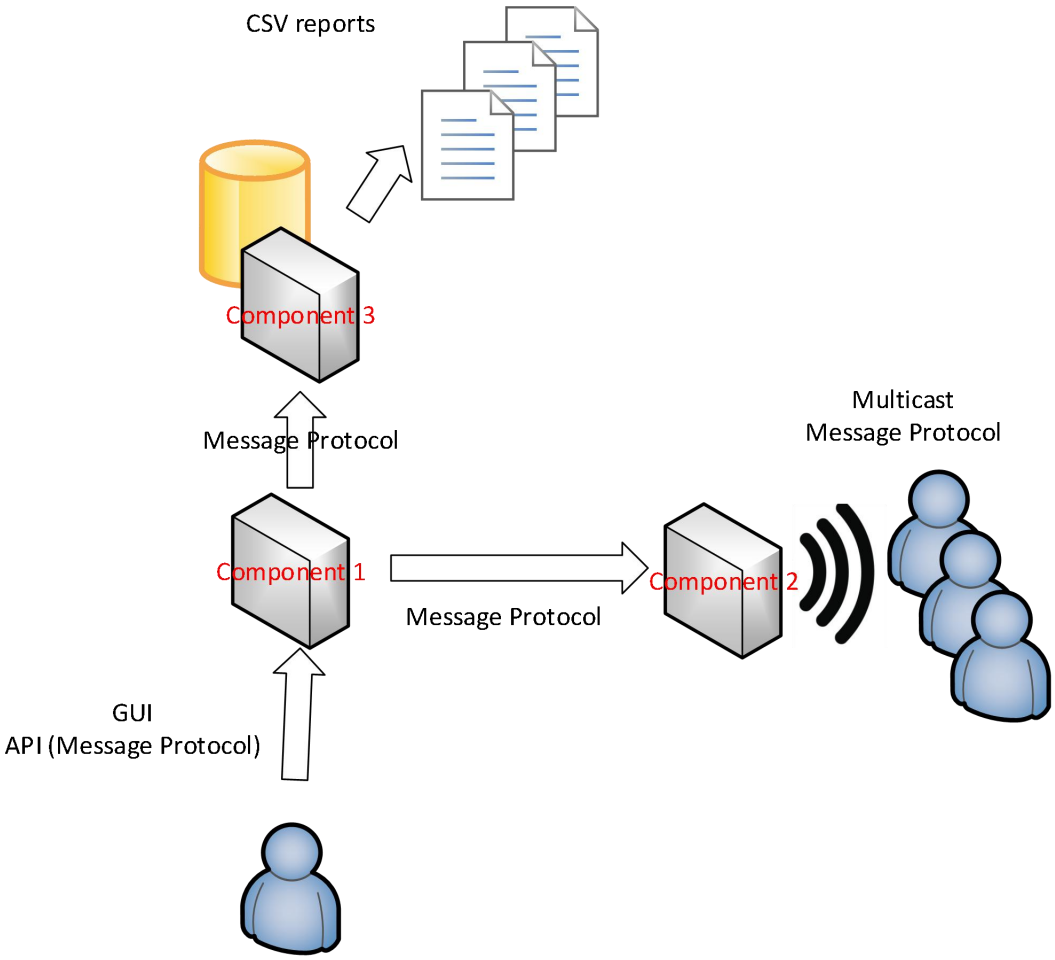
Please answer the following questions in 30 - 45 minutes to the best of your ability.

Test Approach:

The diagram below illustrates a simple IT system comprising three distinct components. Component 1 allows users to input data via GUI or API using a simple name-value message protocol. Component 1 removes any client- and company-specific details from a message before passing it to Component 2 for broadcast to the market. The full data set (with client and company details) flows into component 3, where it is stored in a database. A set of client reports is then generated from the database in CSV format.

Example of a message

Client=Bob Smith|Company=ABC|Product=ISX|Quantity =100|Price=17| ...



A project has been initiated to add a new compulsory name-value pair to API, GUI and the message protocol, across the system. The new name-value pair will contain the Unique Client Identifier, deemed to be client-specific information. The new value will be of fixed length and comprise of 3 letters (Client Mnemonic) followed by 9 digits (example UniqCltID=JPM000023789).

Based on the above information, please outline your approach to designing an automated test strategy for this change. Specifically, address:

1. How would you identify and prioritise the test cases needed across all components?

2. What types of automated tests would you implement at each layer (GUI, API, messaging, database, reporting), and why?

3. How would you ensure both backwards compatibility and data integrity throughout the system after this change?

4. Can you provide an example of a test case you would automate for Component 1?

5. What challenges do you anticipate in automating validation for the new UniqCltID field, and how would you address them?